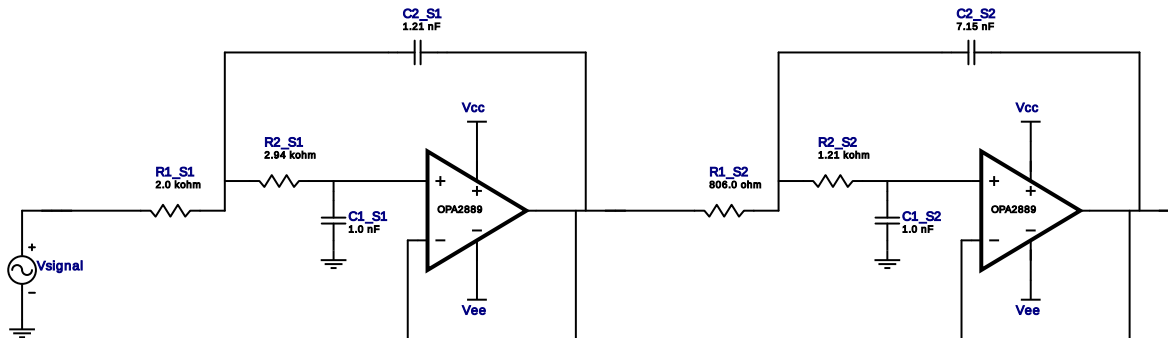


Filter Design Report

Design : Lowpass Filter - 4th order Butterworth
Design ID: 30

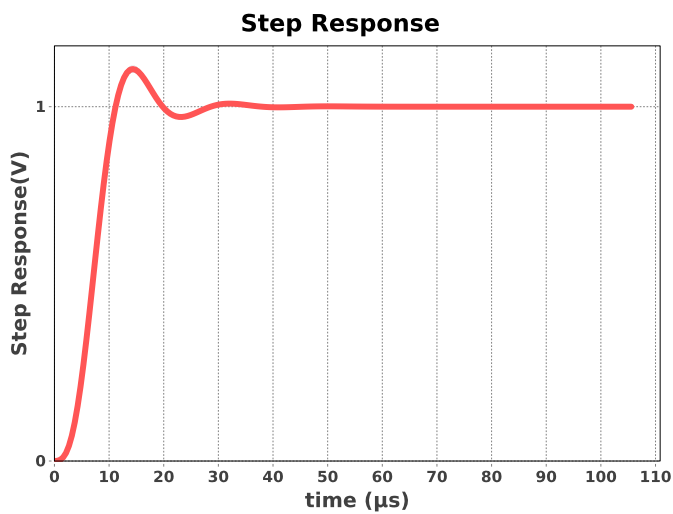
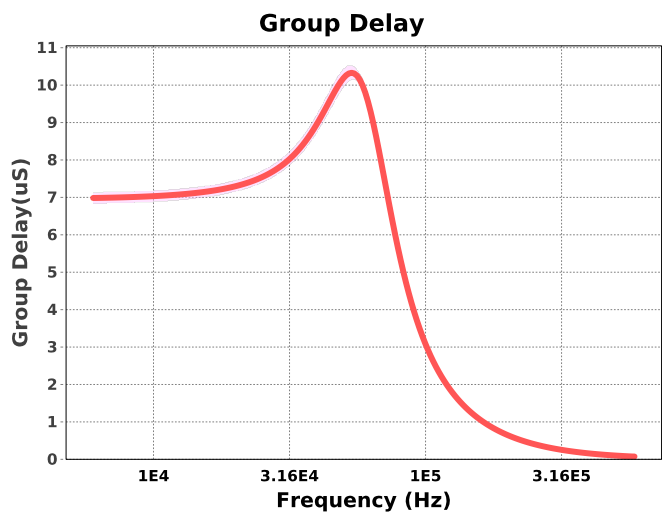
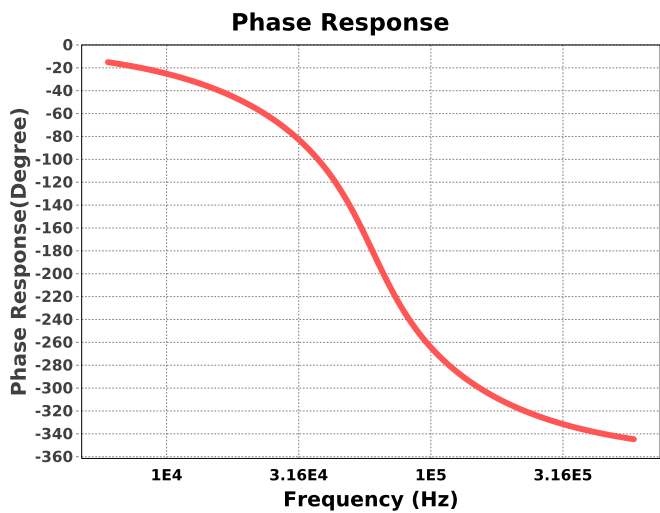
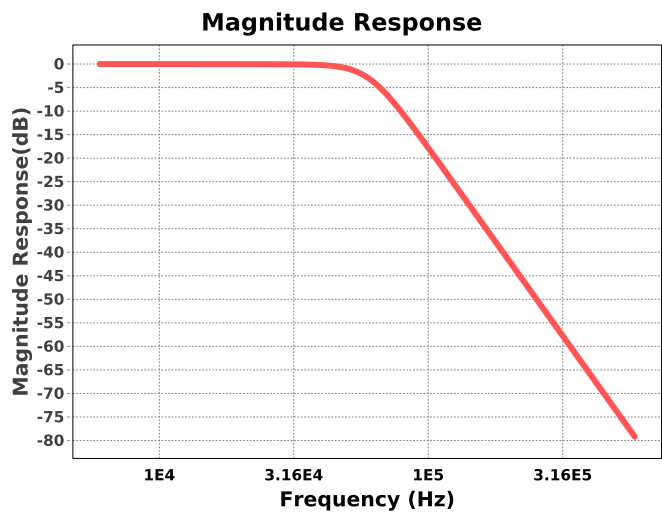


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA2889	GbwTyp= 115MHz VccMax= 12V VccMin= 2.6V	1
2.	A1_S2	Texas Instruments Inc.	OPA2889	GbwTyp= 115MHz VccMax= 12V VccMin= 2.6V	1
3.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
4.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
5.	C2_S1	Generic	Ideal	Cap= 1.21 nF Tolerance= 2.0 %	1
6.	C2_S2	Generic	Ideal	Cap= 7.15 nF Tolerance= 2.0 %	1
7.	R1_S1	Generic	Ideal	Res= 2000.0ohm Tolerance= 1%	1
8.	R1_S2	Generic	Ideal	Res= 806.0ohm Tolerance= 1%	1
9.	R2_S1	Generic	Ideal	Res= 2940.0ohm Tolerance= 1%	1
10.	R2_S2	Generic	Ideal	Res= 1210.0ohm Tolerance= 1%	1

Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	2%
2.	Res	E96	1%



Design Inputs

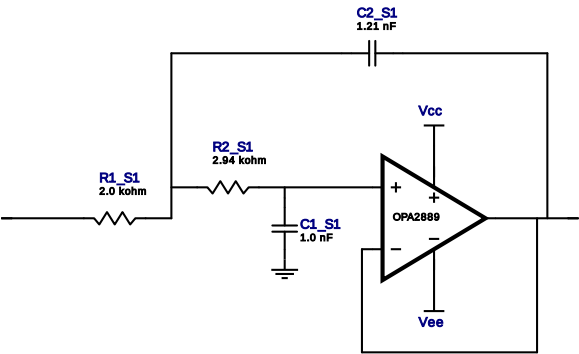
#	Name	Value	Description
1.	FilterType	lowpass	
2.	FilterResponse	Butterworth	
3.	FilterOrder	4.0	
4.	FilterTopology	Sallen-Key	
5.	NumberOfStages	2.0	
6.	PassbandFrequency	60.0 k	
7.	StopbandAttenuation	-73.665	
8.	StopbandFrequency	500.0 k	
9.	Gain	1.0	
10.	DualSupply	+/-5.00 V	Power supply(s) to active chips
11.	ResistorTolerance	E96	Resistor series - 1% Passive resistor tolerance
12.	CapacitorTolerance	E48	Capacitor series - 2% Passive capacitor tolerance

Design Assistance

1. **OPA2889** Product Folder : <http://www.ti.com/product/OPA2889> : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency 59.668 kHz
Min GBW Req'd 3.247 MHz
Stage Gain 1.0 V/V
Stage Q 539.951 m
Stage Topology Sallen-Key

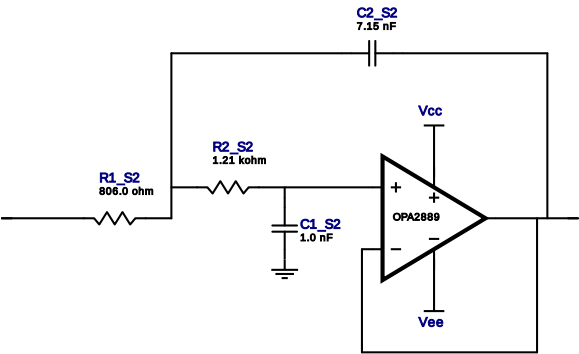


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA2889	GbwTyp= 115MHz VccMax= 12V VccMin= 2.6V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 1.21 nF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 2000.0ohm Tolerance= 1%	1
5.	R2_S1	Generic	Ideal	Res= 2940.0ohm Tolerance= 1%	1

Filter Stage :2

Cutoff Frequency 60.271 kHz
Min GBW Req'd 7.839 MHz
Stage Gain 1.0 V/V
Stage Q 1.31
Stage Topology Sallen-Key



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	OPA2889	GbwTyp= 115MHz VccMax= 12V VccMin= 2.6V	1
2.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 7.15 nF Tolerance= 2.0 %	1
4.	R1_S2	Generic	Ideal	Res= 806.0ohm Tolerance= 1%	1
5.	R2_S2	Generic	Ideal	Res= 1210.0ohm Tolerance= 1%	1

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